



Iranian Mathematical Society Competition

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by mojyla222

– Day 1

P1 A jar contains 1404 white marbles and 1403 black marbles. We randomly remove two marbles from the jar. If the two marbles are the same color, we discard them and add one white marble to the jar. If the two marbles are of different colors, we discard them and add one black marble to the jar. We repeat this process until only one marble remains in the jar. What is the color of the last marble left in the jar? Prove your claim.

P2 Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that for all $x \in [0, 1]$,

$$1404f(x) = f\left(\frac{x}{1404}\right) + f\left(\frac{x+1}{1404}\right) + \cdots + f\left(\frac{x+1403}{1404}\right).$$

Prove that f is a constant function.

P3 Let A be a nonzero $n \times n$ real matrix such that $A = AJA$, where J is the $n \times n$ matrix with all entries equal to 1. Prove that the matrix $A + J$ is not idempotent.

P4 Let m be a natural number. We start with an empty jar and a large number of balls. Tara fills the jar with balls in the following way: If the number of balls in the jar is less than m , she chooses one of the numbers 1, 2, or 3 with equal probability, independently of previous steps, and adds the chosen number of balls to the jar. This process stops when the jar contains at least m balls. Let p_m be the probability that at the stopping time, the jar contains exactly m balls. Compute $\lim_{m \rightarrow \infty} p_m$.

P5 Let (X, d) be a compact metric space with no isolated points. Prove that there exists a finite subset $S \subseteq X$ such that for every $x \in S$, the set $\{d(x, y) : y \in S\}$ contains at least 1404 distinct real numbers.

P6 Let G be a group such that G' is finite. Prove that there exists a finite group H such that $G' \cong H'$.

– Day 2

P7 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that for every $x \in \mathbb{R}$, we have $f(x)f(x^2) \leq 0$. Prove that $f'(0) = 0$.

P8 Let G be a group and H a proper subgroup of G . Suppose there exists a natural number n such that the order of every element of G not in H is equal to n . Prove that n is a power of a prime number.

P9 Let $a_1, a_2, \dots$ be a sequence of natural numbers with the following properties:
 $a_1 = 1$ and for every $i \in \mathbb{N}$,

$$0 \leq a_{2i} - a_{2i-1} \leq 1404,$$

$$a_{2i+1} = \varphi(a_{2i}).$$

Prove that this sequence is bounded.

P10 Let R be a non-trivial ring admitting identity element and $Z(R)$ its center. Suppose that for every $0 \neq r \in R$, there exist $z \in Z(R)$ such that $0 \neq zr \in Z(R)$. Prove that there is an Ideal M in R such that $\frac{R}{M}$ is a field.

P11 Let $f, g : \mathbb{C} \rightarrow \mathbb{C}$ be analytic functions such that for every $z_0 \in \mathbb{C}$,

$$\{f(z) : z \in \mathbb{C}, |z - z_0| \leq 1\} = \{g(z) : z \in \mathbb{C}, |z - z_0| \leq 1\}.$$

Prove that for every $z \in \mathbb{C}$, $f(z) = g(z)$.

P12 Let n be a natural number, and let F be a family of 11-element subsets of $\{1, 2, \dots, n\}$ such that any two members of F has a non empty intersection. A subset of $\{1, 2, \dots, n\}$ is called a *hitting set* for F if it has a non-empty intersection with every member of F . Suppose the minimum size of a hitting set for F is 5. A family H of subsets $B \in H$ is called a *super-hitting set* for F if for every $A \in F$, there exists a member $B \in H$ such that $B \subseteq A$. Prove that there exists a super-hitting set H for F where each member of H has exactly 5 elements, and $|H| \leq 75000$.